Algebraic Number Theory NMAL431
Exercise Sheet 3

3 November 2026

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1. Let K be a number field, and let E/K be a finite Galois extension. Let $\mathfrak{p}$ be a non-zero prime ideal in $\mathcal{O}_K$, and let $\mathfrak{q}$ be a non-zero prime ideal in $\mathcal{O}_E$ lying above $\mathfrak{p}$.

   Let $E^D \subseteq E$ be the fixed field of the decomposition group $D := D_{\mathfrak{q}/\mathfrak{p}}$, and let $E^I \subseteq E$ be the fixed field of the inertia group $I := I_{\mathfrak{q}/\mathfrak{p}}$. Let $\mathfrak{q}^D := \mathfrak{q} \cap E^D$ and let $\mathfrak{q}^I := \mathfrak{q} \cap E^I$.

   This therefore gives us a tower of extensions $E/E^I/E^D/K$ corresponding to the tower of primes $\mathfrak{q}/\mathfrak{q}^I/\mathfrak{q}^D/\mathfrak{p}$.

   (a) Prove that $\mathfrak{q}^I$ is totally ramified in E/E^I (i.e. $e_{\mathfrak{q}/\mathfrak{q}^I} = [E : E^I]$).

   (b) Prove that $\mathfrak{q}^D$ is totally inert in E^I/E^D (i.e. $f_{\mathfrak{q}^I/\mathfrak{q}^D} = [E^I : E^D]$).

   (c) Hence or otherwise, deduce that $\mathfrak{q}$ is the unique prime ideal of $\mathcal{O}_E$ lying above $\mathfrak{q}^D$.

2. Let K be a number field, and let E/K be a finite Galois extension. Let $\mathfrak{p}$ be a non-zero prime ideal in $\mathcal{O}_K$, and let $\mathfrak{q}$ be a non-zero prime ideal in $\mathcal{O}_E$ lying above $\mathfrak{p}$. As before, let $E^D \subseteq E$ and $E^I \subseteq E$ denote the fixed fields of $D_{\mathfrak{q}/\mathfrak{p}}$ and $I_{\mathfrak{q}/\mathfrak{p}}$ respectively.

   Let L be a number field such that $K \subseteq L \subseteq E$, and let $\mathfrak{q}_L := \mathfrak{q} \cap L$.

   (a) Prove that if $e_{\mathfrak{q}_L/\mathfrak{p}} = 1$ and $f_{\mathfrak{q}_L/\mathfrak{p}} = 1$, then $L \subseteq E^D$.

   (b) Prove that if $e_{\mathfrak{q}_L/\mathfrak{p}} = 1$, then $L \subseteq E^I$.

   (c) Prove that if $\mathfrak{q}$ is the only prime of $\mathcal{O}_E$ lying above $\mathfrak{q}_L$, then $E^D \subseteq L$.

3. Let K be a number field, and let L_1/K and L_2/K be two finite extensions of K, lying inside a fixed algebraic closure of K.

   (a) Let $\mathfrak{p}$ be a non-zero prime ideal of $\mathcal{O}_K$ which is unramified in L_1 and L_2. Prove that $\mathfrak{p}$ is unramified in the compositum $L_1 \cdot L_2$.

   (b) Let $\mathfrak{p}$ be a non-zero prime ideal of $\mathcal{O}_K$ which splits completely in L_1 and L_2. Prove that $\mathfrak{p}$ splits completely in the compositum $L_1 \cdot L_2$.

[*Hint: Let E be the Galois closure of $(L_1 \cdot L_2)/K$ and apply the previous exercise*]

4. (*The Dedekind-Kummer theorem*) Let A be a Dedekind domain with fraction field K, E/K a finite separable extension, and B the integral closure of A in E. Let $\alpha \in B$ such that $E = K(\alpha)$ and let $f(X) \in A[X]$ be the minimal polynomial of α.

   Let $\mathfrak{p}$ be a non-zero prime ideal of A, and suppose that $\overline{f}$ factorises as $\prod_{i=1}^{r} \overline{g_i}^{e_i}$ in $k_\mathfrak{p}[X]$ with the $\overline{g}_i$ distinct, monic and irreducible, and let $g_i \in A[X]$ be monic lifts.

   (a) Suppose that $A_\mathfrak{p}[\alpha] = B_\mathfrak{p}$. Show that $B/\mathfrak{p}B \cong k_\mathfrak{p}[X]/(\overline{f})$.
   [*Hint: Localise at $\mathfrak{p}$ and use $A/\mathfrak{p} \cong A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$*]

   (b) Deduce that $\mathfrak{q}_i = \mathfrak{p}B + g_i(\alpha)B$ are the distinct primes of B lying above $\mathfrak{p}$, and that $\mathfrak{p}B = \prod_i \mathfrak{q}_i^{e_i}$ and that $f_{\mathfrak{q}_i/\mathfrak{p}} = \deg \overline{g}_i$
   [*Hint: Use the Chinese remainder theorem in $k_\mathfrak{p}[X]/(\overline{f})$*]

   (c) Show that $A_\mathfrak{p}[\alpha] = B_\mathfrak{p}$ for all but finitely many prime ideals $\mathfrak{p}$. In the case where $A = \mathbb{Z}$, show that $A_\mathfrak{p}[\alpha] = B_\mathfrak{p}$ if and only if $p \nmid [\mathcal{O}_E : \mathbb{Z}[\alpha]]$.

5. Let $K = \mathbb{Q}(\sqrt[3]{2})$.

   (a) Show that 2 and 3 are totally ramified in K, and that no other prime ramifies in K.

   (b) Let $p \equiv 2 \bmod 3$ with $p \neq 2$. Show that $p\mathcal{O}_K = \mathfrak{q}_1\mathfrak{q}_2$ with $f_{\mathfrak{q}_1/\mathfrak{p}} = 1$ and $f_{\mathfrak{q}_2/\mathfrak{p}} = 2$.

   (c) Let $p \equiv 1 \bmod 3$. Show that p splits completely in K if 2 is a cube modulo p, and is inert otherwise. Find the two smallest primes which split completely.

   [*Hint: You may use that $\mathcal{O}_K = \mathbb{Z}[\sqrt[3]{2}]$*]

6. Can you find an example of a number field $K/\mathbb{Q}$ such that, for every rational prime p, there are an even number of prime ideals $\mathfrak{p}$ in $\mathcal{O}_K$ lying above p ?

   [*Hint: Try a biquadratic field; i.e. $\mathbb{Q}(\sqrt{D_1}, \sqrt{D_2})$ for some suitable $D_1, D_2 \in \mathbb{Z}$*]